CHAPTER 09

Time, Speed and Distance

It is sure and certain that it is one of the most important chapters, which includes various fundamental and logical concepts and therefore most of the problems are complex in their nature. Every year 4-6 problems are generally asked in CAT. Besides CAT almost every aptitude exam contains the questions pertaining to the concepts of TSD.

This chapter includes the following :

(a) Motion in a straight line (b) Circular motion and races

(c) Problems based on trains, boats, river and clocks etc.

9.1 Concept of Motion

When a body moves from a point A to another point B at a distance of D, then it requires some time (T) to cover a distance (D) with a particular speed (S).

The relation between T, S and D is as follows : $T \times S = D$ *i.e.*, Time × Speed = Distance

Therefore, when D is constant, $T \propto \frac{1}{S}$ and when T is constant, $D \propto S$ and when S is constant, $D \propto T$

NOTE This relation of proportionality is very important.

Formulae : $\text{Distance} = \text{Speed} \times \text{Time}$, $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$ or $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

NOTE To solve the problem, all the units involved in the calculation must be uniform *i.e.,* either all of them be in metres and seconds or in kilometres and hours etc.

Conversion of unit: $1 \text{ km/h} = \frac{5}{18} \text{ m/s}$

$$1 \text{ m/s} = \frac{18}{5} \text{ km/h} \; [1 \text{ km} = 1000 \text{m}, \quad 1 \text{ h} = 60 \text{ min}, \quad 1 \text{ min} = 60 \text{ s}]$$

Hint Try to find these relations by unitary method.

$$1 \text{ mile} = 1609.30 \text{ m} = 1.6093 \text{ km}$$

and $1 \text{ km} = 0.621 \text{ mile}, 1 \text{ yard} = 0.9144 \text{ m}$

Chapter Checklist

- Concept of Motion
- Relative Motion with Two or More Bodies
- To and Fro Motion in a Straight Line
- Concept Based on Trains
- Concept Based on Boats and Rivers (or streams)
- Races
- Circular Motion
- Clocks
- **CAT Test**

and $1\text{ m} = 1.0936$ yards and $1\text{ m} = 39.4$ inches

$$\textbf{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}}$$

When distances are equal :

$$\text{Average speed} = \frac{2xy}{x+y} \qquad (\text{speeds } x,\ y)$$

and $$\text{Average speed} = \frac{3xyz}{xy+yz+zx} \qquad (\text{speeds } x,\ y,\ z)$$

Exp. 1) Abhishek drives his bike at the speed of 150 km/h. What is the distance covered by him in 3 hours.

Solution $D = S \times T = 150 \times 3 = 450$ km

Exp. 2) Udai travels half of his journey by train at the speed of 120 km/h and rest half by car at 80 km/h. What is the average speed?

Solution Let the total distance be $2D$ km, then

Total time = Time taken by train + Time taken by car $= \frac{D}{120} + \frac{D}{80}$

$$\therefore \quad \text{Average speed} = \frac{2D}{\frac{D}{120}+\frac{D}{80}} = \frac{2D}{D\left(\frac{1}{120}+\frac{1}{80}\right)} = \frac{2}{\left(\frac{2+3}{240}\right)} = 96$$

km/h

Alternatively Consider the distance in numerals (since it is independent of the distance) as the LCM of the speeds.

Then, the total distance = LCM of 120 and 80 = 240

Therefore, time taken by train $= \frac{120}{120} = 1$ h

and time taken by car $= \frac{120}{80} = \frac{3}{2}$ h

$$\therefore \quad \text{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}} = \frac{240}{1+\frac{3}{2}} = 96 \text{ km/h}$$

Alternatively Average speed $= \frac{2(xy)}{(x+y)}$

where x and y are the speeds.

$$\therefore \quad \text{Average speed} = \frac{2 \times 120 \times 80}{120+80} = 96 \text{ km/h}$$

NOTE This formula is applicable only when the distances travelled at the speed of x km/h and y km/h are same.

Alternatively Solve through alligations.

Exp. 3) Speed of Karan is 40 km/h and speed of Arjun is 60 km/h. What is the ratio of time taken by each to cover the same distance?

Solution	**Karan**		**Arjun**
Speed	40	:	60
$\Rightarrow$	2	:	3
$\therefore$ Time	3	:	2

Alternatively Time taken $= \frac{1}{40} : \frac{1}{60} = 3 : 2$

NOTE For the constant distance covered, the ratio of time taken by each person is the ratio of the reciprocal of their speeds and *vice-versa.*

If the ratio of speeds of A, B, C, D be $a : b : c : d$, then the ratio of time taken by A, B, C and D to cover the same distance

$$= \frac{1}{a} : \frac{1}{b} : \frac{1}{c} : \frac{1}{d}$$

and if the time taken by A, B, C and D be $p : q : r : s$, then the ratio of their speeds $= \frac{1}{p} : \frac{1}{q} : \frac{1}{r} : \frac{1}{s}$

Exp. 4) A car takes half of the time taken by truck to go from Lucknow to Bombay. A truck takes 20 hours to go for the same journey. What is the speed of truck, if the speed of car be 120 km/h.

Solution	Car		Truck
Time $\rightarrow$	1	:	2
Speed $\rightarrow$	2	:	1

Therefore, speed of the truck = 60 km/h

Exp. 5) A cycle covers 75 km distance in 5 hours. What is the distance covered by the cycle in 6 hours?

Solution $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

$$\therefore \quad \text{Speed} = \frac{75}{5} = 15 \text{ km/h}$$

Now, distance covered in 6 hours at the speed of 15 km/h

$$= 15 \times 6 = 90 \text{ km}$$

Exp. 6) The distance of the college and the home of Rajeev is 80 km. One day he was late by 1 hour than the normal time to leave for the college, so he increased his speed by 4 km/h and thus he reached to college at the normal time. What is the changed (or increased) speed of Rajeev?

(a) 28 km/h (b) 30 km/h (c) 40 km/h (d) 20 km/h

Solution Let the normal speed be x km/h, then

$$\frac{80}{x} - \frac{80}{(x+4)} = 1$$

$$\Rightarrow \quad x^2 + 4x - 320 = 0$$

$$\Rightarrow \quad x(x+20) - 16(x+20) = 0$$

$$(x+20)(x-16) = 0$$

$$x = 16 \text{ km/h}$$

$$\therefore \quad (x+4) = 20 \text{ km/h}$$

Therefore increased speed = 20 km/h

Alternatively If t_1 and t_2 be the original and changed time and S_1 and S_2 be the original and changed speeds, then

$$S_1 \times S_2 = \frac{\text{Distance} \times (S_1 \sim S_2)}{(t_1 \sim t_2)}$$

Also, $$\boxed{t_1 \times t_2 = \frac{\text{Distance} \times (t_1 \sim t_2)}{(S_1 \sim S_2)}}$$

$$\therefore \quad S_1 \times S_2 = \frac{80}{1} \times 4 = 320 \quad \therefore \quad S_1 \times S_2 = 320$$

or $S_1 \times (S_1 + 4) = 320 \Rightarrow S_1 = 16$ and $S_2 = 20$

NOTE Here you need not to solve necessarily quadratic equation. You can just try and find two factors of 320 in such a way that (in this particular problem) one factor must be greater than the other factor by 4.

Alternatively Go through options. Consider 20

$$t_2 = \frac{80}{20} = 4 \Rightarrow t_1 = 4 + 1 = 5$$

$$\therefore \quad S_1 = \frac{80}{5} = 16$$

Since, S_1 is 4 km/h less than S_2. Hence, option (d) is correct.

Alternatively If you find yourself comfortable then factorise 80 in two factors and check the correct combinations.

$$80 = \begin{matrix} \textcircled{-1} \\ \left[\begin{matrix} 1 \times 80 \\ 2 \times 40 \\ 4 \times 20 \\ 5 \times 16 \\ 8 \times 10 \end{matrix}\right] \\ \end{matrix} \begin{matrix} \\ \\ \\ \\ \\ \end{matrix} +\textcircled{4}$$

So, the increased speed = 20 km/h.

Exp.7) Shweta when increases her speed from 24 km/h to 30 km/h she takes one hour less than the usual time to cover a certain distance. What is the distance usually covered by Shweta?

(a) 160 km (b)240 km (c) 120 km (d) 90 km

Solution Let the original time be t hours, then

$$24t = 30 \times (t - 1) = D \qquad \text{(distance)}$$

Then $t = 5$

$$\therefore \quad \text{Distance} = 24 \times 5 = 120 \text{ km}$$

Alternatively Go through options.

$$\left.\begin{matrix} \frac{120}{24} = 5 \text{ h} \\ \frac{120}{30} = 4 \text{ h} \end{matrix}\right\} \text{ 1 hour less}$$

Hence, the option (c) is correct.

Alternatively Since distance (D) is constant.

Therefore, $D = S_1 \times t_1 = S_2 \times t_2$

It means here we can apply product constancy

Speed	Time	
$\frac{1}{4}\uparrow$	$\frac{1}{5}\downarrow$	$= 1$ hour

$\therefore$ Original time taken $= 5 \times 1 = 5$ hours

Therefore, Distance = Original speed × Original time

$$= 24 \times 5 = 120 \text{ km}$$

NOTE In the given exercise or in the whole chapter you have to solve maximum problems through product constancy concept described in the chapter of ratio proportion and variation.
Solving through product constancy gives faster results.

Alternatively Let the distance be D, then

$$\frac{D}{24} - \frac{D}{30} = 1 \Rightarrow D = 120 \text{ km}$$

Exp. 8) Kriplani goes to school at 20 km/h and reaches the school 4 minutes late. Next time, she goes at 25 km/h and reaches the school 2 minutes earlier than the scheduled time. What is the distance of her school?

Solution Increase in speed = 5 km/h

Decrease in time = 6 min (4 + 2)

By product constancy :

Speed	Time	
$\frac{1}{4}\uparrow$	$\frac{1}{5}\downarrow$	$= 6$ min

It means original time = 30 min $\left(\because \frac{x}{5} = 6 \Rightarrow x = 30\right)$

$\therefore$ Total distance = Original speed × Original time

$$= 20 \times \frac{30}{60} = 10 \text{ km}$$

Exp. 9) Amit covers a certain distance with his own speed, but when he reduces his speed by 10 km/h his time duration for the journey increases by 40 hours, while if he increases his speed by 5 km/h from his original speed he takes 10 hours less than the original time taken. Find the distance covered by him.

Solution

$$\left.\begin{matrix} S & \times & T \\ -10 & & +40 \end{matrix}\right\} \quad 40S - 10T = |-400| \qquad \ldots(i)$$

$$\left.\begin{matrix} S & \times & T \\ +5 & & -10 \end{matrix}\right\} \quad -10S + 5T = |-50| \qquad \ldots(ii)$$

Solving eq. (i) and (ii), we get

$$S = 25 \quad \text{and} \quad T = 60$$

$\therefore$ Distance (D) $= S \times T = 25 \times 60 = 1500$ km

where $D \rightarrow$ Distance, $S \rightarrow$ Speed, $T \rightarrow$ Time

'+' means increase in value.

and '–' means decrease in value.

Alternatively Let distance be x km and usual speed be y km/h.

$$\frac{x}{(y-10)} - \frac{x}{y} = 40$$

$$\Rightarrow \quad x\left[\frac{10}{y(y-10)}\right] = 40$$

$$\Rightarrow \quad x = 4y\,(y - 10) \qquad \ldots(i)$$

and $$\frac{x}{y} - \frac{x}{(y+5)} = 10$$

$$\Rightarrow \quad x = 2y(y + 5) \qquad \ldots(ii)$$

from eq. (i) and (ii)

$$4y(y - 10) = 2y(y + 5)$$

$$2y - 20 = y + 5$$

$$y = 25 \text{ km/h}$$

$$\therefore \quad x = 1500 \text{ km}$$

Exp. 10) A train met with an accident 60 km away from Anantpur station. It completed the remaining journey at $\frac{5}{6}$th of the previous speed and reached the Baramula station 1 hour 12 min late. Had the accident taken place 60 km further, it would have been only 1 hour late.

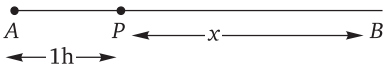
(a) What is the normal speed of the train?

(b) What is the distance between Anantpur and Baramula?

Solution

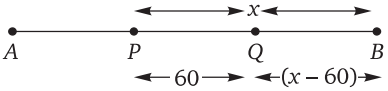
***Case* I.** Since the speed is decreased by $\frac{1}{6}$. So, the time will be increased by 1/5, which is equal to 1 hour 12 minutes. It means the normal time required for this remaining part (x) of the journey is 5×72 min = 360 min = 6 h.

($\because$ 1 h 12 min = 72 min)



P is the place of accident.

***Case* II.** When accident is supposed to be happened at Q.



Since, the speed is decreased by $\frac{1}{6}$, hence, the time will be increased by $\frac{1}{5}$, which is equal to 1 hour, hence the normal time required for this remaining part $(x - 60)$ of journey $= 5 \times 1 = 5$ hours. Thus, it is clear that when the train runs 60 km of its normal speed it takes 1 hour less, which implies that in 1 hour the train can run 60 km with its normal speed. Thus, the normal speed of the train is 60 km/h.

(b) Since the train requires 6 hours at its normal speed of 60 km/h for the x km. Hence, $x = 6 \times 60 = 360$ km

Thus, the total distance = Distance travelled before accident + Distance travelled after accident

$= 60 \times 1 + 60 \times 6 = 420$ km

Exp. 11) A and B started simultaneously towards each other from P and Q respectively. The distance between P and Q is 600 km and the ratio of speeds of A is to B is 5 : 7. If they meet at a point M:

(i) Find the ratio of PM : QM. (ii) Find the distance PM.

Solution

(i) Since the time is constant so, the distance covered by A and B is directly proportional to the speeds of A and B. Hence, $PM : QM = 5 : 7$

(ii) Since, the ratio of their speeds (or their distances covered) is 5 : 7. Hence, A will cover $\frac{5}{12}$ of the total length.

$$\therefore \quad PM = \frac{5}{12} \times 600 = 250 \text{ km}$$

(Similarly $QM = \frac{7}{12} \times 600 = 350$ km)

9.2 Relative Motion with Two or More Bodies

(i) **When two bodies move in the same direction:** If the speeds of the two bodies A and B be S_A and S_B, then their relative speed $= S_A - S_B$ or $S_B - S_A$ *i.e.*, in the same direction the relative speed or effective speed between two bodies is the difference of their speeds. (The difference is always considered as positive)

(ii) **When two bodies move in the opposite direction:** If the speeds of the two bodies A and B be S_A and S_B, then their relative speed $= S_A + S_B$.

i.e., in the opposite direction the relative speed or effective speed between two bodies is the sum of their speeds.

Exp. 12) The distance between two places P and Q is 700 km. Two persons A and B started towards Q and P from P and Q simultaneously. The speed of A is 30 km/h and speed of B is 40 km/h. They meet at a point M which lies on the way from P to Q.

(i) How long will they take to meet each other at M?

(ii) What is the ratio of $PM : MQ$?

(iii) What is the distance MQ?

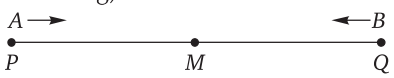
(iv) What is the extra time needed by A to reach at Q than to reach at P by B?

(v) What is the ratio of time taken by A and B to reach their respective destinations after meeting at M?

(vi) In how many hours will they be separated by only 560 km before meeting each other.

(vii) How long will it take to separate them by 280 km from each other when they cross M (time to be considered after their meeting)?

Solution



(i) Since, they are coming towards each other from opposite ends, therefore the relative speed will be the sum of their speeds = 30 + 40 = 70 km/h.

Thus, the required time to meet at M

= Time required to cover 700 km (combined)

$$= \frac{700}{70} = 10 \text{ h}$$

Thus in 10 hours they will meet each other at M.

(ii) The ratio of their distances covered to meet at M

= Ratio of their speeds = 3 : 4

(Since, time is constant *i.e.*, same for each)

Thus $PM : MQ = 3 : 4$

(iii) $MQ = \frac{4}{7} \times 700 = 400$ km

(iv) Time required by A to reach at $Q = \frac{700}{30} = \frac{70}{3}$ h

Time required by B to reach at $P = \frac{700}{40} = \frac{70}{4}$ h

$\therefore$ Extra time required by $A = \frac{70}{3} - \frac{70}{4}$

$= 70 \times \frac{1}{12} = 5$ h 50 min

(v) Time required by A to cover $MQ = \frac{400}{30}$

and time required by B to cover $MP = \frac{300}{40}$

$\therefore$ Required ratio $= \frac{400/30}{300/40} = \frac{16}{9}$

Remember If speed of A is S_A and speed of B is S_B and A takes t_A time to cover MQ and B takes t_B time to cover MP, then $\frac{S_A}{S_B} = \sqrt{\frac{t_B}{t_A}}$

(vi) It means they have to cover $(700 - 560) = 140$ km. Thus, the required time to cover 140 km distance $= \frac{140}{70} = 2$ h

(vii) Since in each hour they separate by 70 km from each other. Hence, to separate by 280 km, time required

$= \frac{280}{70} = 4$ h

Exp. 13) Two places P and Q are 800 km apart from each other. Two persons start from P towards Q at an interval of 2 hours. Whereas A leaves P for Q before B. The speeds of A and B are 40 km/h and 60 km/h respectively. B overtakes (or catches or meets) A at M, which is on the way from P to Q.

(i) How long will B take to overtake?

(ii) What is the distance from P, where B overtakes A (i.e., PM)?

(iii) What is the ratio of time taken by A and B to meet at M?

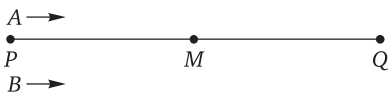
(iv) What is the extra time required by A to reach at Q?

(v) How many hours late A will reach at Q than that of B?

(vi) After how many hours A and B will be separated by 50 km before M, when both are moving?

(vii) How many hours does B require to advance himself, by 100 km, in comparison to A?

Solution



(i) Since A moves 2 hours earlier than B at the speed of 40 km/h, so in 2 hours A will cover 80 km. Thus A will be 80 km away from B.

Now, since they are moving in the same direction therefore their relative speed will be $(60 - 40) = 20$ km/h *i.e.*, with respect to A, B moves at 20 km/h. It means either they reduce (or create) the difference of 20 km in each hour between themselves.

Thus, the required time to overtake

(or reduce the difference upto zero from 80 km)

$= \frac{\text{Distance advanced}}{\text{Relative speed}} = \frac{80}{20} = 4$ h

(ii) The distance between P and M

= (Time required to overtake × Speed of the faster body)

$= 4 \times 60 = 240$ km

Since, B has to move for 4 hours at 60 km/h. Hence, distance covered = 240 km.

(iii) Time taken by B to reach at $M = 4$ h

Time taken by A to reach at $M = (4 + 2) = 6$ h

Thus, the ratio of time taken by A and B $= 6 : 4 = 3 : 2$

(Since, A has left 2 hours earlier)

(iv) Time taken by A to reach at $Q = \frac{800}{40} = 20$ h

Time taken by B to reach at $Q = \frac{800}{60} = 13$ h 20 min

Hence, A takes 6 hours 40 min extra to reach at Q.

(v) Since, A leaves 2 hours earlier, thus he will reach at Q only 4 hours 40 min late.

Since, A takes 6 hours 40 min extra to reach at Q.

(vi) When B starts to move towards Q the difference between A and B is 80 km.

The required difference between A and $B = 50$ km

Hence, they have to reduce it by 30 km.

Thus, the required time $= \frac{(80 - 50)}{20} = 1.5$ h $= 1$ h 30 min

Thus, after 3/2 hours A will be only 50 metres ahead of B.

(vii) When B starts to follow A (towards Q) A was 80 km ahead of B. Also B wants to overtake A and further go ahead of A by 100 km. Thus, the net difference (required) = 180 km.

$\therefore$ Required time

$= \frac{(\text{Distance advanced + Required difference})}{\text{Relative speed}}$

$= \frac{80 + 100}{20} = \frac{180}{20} = 9$ h

Thus, after 9 hrs (when B starts moving) B will be 100 km ahead of A *i.e.*, they will be separated by 100 km from each other after crossing M.

9.3 To and Fro Motion in a Straight Line

This concept is just the extension of the previous concepts of relative motion between more than one dynamic (or moving) bodies.

(a) *When two bodies start moving towards each other*

Let the initial distance (or gap) between two bodies A and B be D, and their respective speeds be S_A and S_B, then A and B together have to cover D unit of distance for the first meeting, irrespective of their speeds.

(i) To meet each other they cover the distances in the ratio of their individual speeds.

(ii) If the initial distance (or gaping) between two bodies A and B is D, then A and B together have to cover D unit of distance for the first meeting.

(iii) For the next number of meeting (*e.g.,* second, third, fourth meeting and so on) both A and B together have to cover $2D$ distance more from the previous meeting *i.e.,* to meet for the fourth time they have to cover together $D + (3 \times 2D) = 7D$ unit of distances. Similarly for seventh meeting they have to cover together $D + (6 \times 2D) = 13D$ units of distance. Thus for each subsequent meeting they have to cover $2D$ distance extra from the previous one.

NOTE Individually they will cover the distances in the ratio of their speeds for any number of meeting. Thus, the total distance covered for the nth meeting $= (2n - 1)\,D$.

(b) *When two bodies start moving towards the same direction*

Let the two bodies A and B be initially at the same end of the track, their respective speeds be S_A and S_B and D be the length of the track, then A and B together have to cover $2D$ unit of distance for the first meeting, irrespective of their speeds.

(i) For the first meeting after they start to move they have to cover $2D$ distance, if the distance between two particular points (or places) be D unit. Since, the faster body reaches the next (or opposite) end first than the slower body and the faster body starts returning before the slower body reaches the same opposite end and thus the two bodies meet somewhere between the two ends covering individually the distances in their respective speeds.

(ii) For every subsequent meeting they have to cover together $2D$ unit distance more from the previous meeting.
Thus, for nth meeting they have to cover together $(n \times 2D)$ unit of distance.

(iii) At any point of time the distances covered by the bodies will be equal to the ratio of their speeds.

Exp. 14) The distance between P and Q is 100 m and the speeds of A and B are 20 m/s and 30 m/s respectively. Initially A and B are at P. They move between P and Q. Calculate:

(i) The distance covered by A in first meeting. (ii) The time required for the second meeting.
(iii) The distance covered by B to meet for the third time. (iv) The ratio of distances covered by A and B till the fourth meeting.
(v) The distance between P and the place of fifth meeting.
(vi) The distance between Q and the point of third meeting.

Solution

Meeting	Total distance covered by A and B together	Distance covered by A	Distance covered by B	Point of meeting from P	Point of meeting from Q	Time (in second)
1st	200 m	80 m	120 m	80 m	20 m	4
2nd	400 m	160 m	240 m	40 m	60 m	8
3rd	600 m	240 m	360 m	40 m	60 m	12
4th	800 m	320 m	480 m	80 m	20 m	16
5th	1000 m	400 m	600 m	0 m (at P)	100 m	20

(i) The distance covered by A in the first meeting = 80 m (ii) Time required for the second meeting = 400/50 = 8 s
(iii) The distance covered by B for the third time meeting = 12 × 30 = 360 m
(iv) It is always (for any moment, after the starting of movement) will be in the ratio of their respective speeds.
So, the required ratio of distances covered by A and B = 20 : 30 = 2 : 3 *i.e.,* $A : B$ = 20 : 30 = 2 : 3
(v) Since for the fifth meeting they have to cover 50 × 20 = 1000 m.
(vi) 60 m.

Exp. 15) The distance between two points P and Q is 100 m. A is initially at P and B is at Q. The speeds of A and B is 20 m/s and 30 m/s. They move between P and Q to and fro :

(i) Find the time required for the first meeting. (ii) Distance covered by A till the third meeting.
(iii) Distance covered by B till the fifth meeting. (iv) The distance between P and the place of fourth meeting.
(v) The distance between Q and the place of fifth meeting. (vi) The ratio of distances covered by each one till the third meeting.

Solution

No. of meeting	Distance covered by A and B together	Time (in second)	Distance covered by A	Distance covered by B	Distance between P and point of meeting	Distance between Q and point of meeting
1st	100 m	2	40 m	60 m	40 m	60 m
2nd	300 m	6	120 m	180 m	80 m	20 m
3rd	500 m	10	200 m	300 m	at P	at P
4th	700 m	14	280 m	420 m	80 m	20 m
5th	900 m	18	360 m	540 m	40 m	60 m

(i) 2 second $\left(\text{Time} = \frac{100}{50} = 2\right)$

(ii) 200 m (Distance = Speed × Time = $20 \times 10 = 200$ m)

(iii) 540 m ($D = 18 \times 30 = 540$ m)

(iv) 80 m

(v) 60 m

(vi) 2 : 3 always equal to the ratio of respective speeds.

Exp. 16) A and B are two friends. A lives at a place P and B lives at another place Q. Everyday A goes to Q to meet B at 120 km/h. Thus, he takes 3 hours. On a particular day B started to meet A so he moved towards P. On that day A took only 2 hours to meet B on the way instead at Q.

(i) What is the ratio of speeds of A is to B?

(ii) What is the speed of B?

Solution Distance between P and $Q = 120 \times 3 = 360$ km

Let the speed of B be S_B, then

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$2 = \frac{360}{(120 + S_B)}$$

$$\Rightarrow \quad S_B = 60 \text{ km/h}$$

Here, A and B are moving towards each other. So, the relative speed will be the sum of the speeds of A and B both.

Therefore, ratio of speeds of $A : B = 2 : 1$.

(i) 2 : 1 (ii) 60 km/h

NOTE In this case ratio of speeds of

$$A:B = \frac{\text{Actual time required when } B \text{ is also moving}}{\text{Time difference when } B \text{ is also moving}} = \frac{(3-1)}{1} = \frac{2}{1}$$

Exp. 17) A lives at P and B lives at Q. A usually goes to meet B at Q. He covers the distance in 3 hours at 150 km/h. On a particular day B started moving away from A. While A was moving towards Q thus A took total 5 hours to meet B.

(i) What is the speed of B?

(ii) What is the ratio of speeds of $A : B$?

Solution Distance $= 3 \times 150 = 450$ km

$$\text{Time} = \frac{\text{Distance}}{\text{Relative speed}}$$

$$5 = \frac{450}{(150 - S_B)} \qquad (S_B \rightarrow \text{Speed of } B)$$

$$\Rightarrow \quad S_B = 60 \text{ km/h}$$

Ratio of speeds of $A : B = \frac{150}{60} = \frac{5}{2} = 5 : 2$

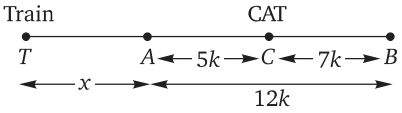
(i) 60 km/h (ii) 5 : 2

In this case ratio of speeds of

$$A : B = \frac{\text{Actual time required when } B \text{ is also moving}}{\text{Time difference}} = \frac{5}{2}$$

Exp. 18) A train approaches a tunnel AB. Inside the tunnel a cat is located at a point which is $\frac{5}{12}$ of the distance AB measured from the entrance A. When the train whistles, the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves to the exit B, the train catches the cat at exactly the exit. The speed of the train is greater than the speed of the cat by what order?

Solution



Let the speed of train be u and the speed of cat be v and train whistles at a point T, x km away from A, then

$$\frac{u}{v} = \frac{x}{5k} = \frac{x + 12k}{7k}$$

$$\Rightarrow \quad 7x = 5(x + 12k)$$

$$\Rightarrow \quad \frac{x}{k} = \frac{30}{1}$$

$$\therefore \quad \frac{u}{v} = \frac{30}{5 \times 1} = \frac{6}{1}$$

Alternatively $\frac{u}{v} = \frac{7k + 5k}{7k - 5k} = \frac{6}{1}$

NOTE Since time is constant, therefore distances covered by train and cat will be in the ratio of their respective speeds.

9.4 Concept Based on Trains

(i) When two trains (or bodies) are moving in opposite direction, their relative speed will be equal to the sum of their individual speeds.

(ii) When two trains are moving in the same direction their relative speed will be equal to the difference of their speeds.

(iii) Distance to be covered to cross each other is always equal to the sum of their individual lengths.

(iv) Distance to be covered such as bridge, platform etc., is always equal to the sum of the length of train and the length of the particular object such as bridge, platform etc.

(v) Distance to be covered such as pole, man, tree etc, is always equal to the length of the train only.

(vi) If a man is travelling in a train, then this man has to cover the distance to cross another train is equal to the length of the train which is passing or crossing him. In this case the relative speed of both the trains will be considered.

Exp. 19) A train crosses a tree in 10 seconds. If the length of the train be 150 m, then find the speed of the train.

Solution Distance = Length of train
= Speed of train × Time
$150 = \text{Speed} \times 10$
$\Rightarrow$ Speed = 15 m/s
$\text{Speed} = 15 \times \frac{18}{5} = 54 \text{ km/h}$

NOTE A train starts to cross a stationary thin object (of inconsiderable thickness) when the engine of the train meets the object and it completes the crossing when the last wagon (or backend) of the train just crosses the object.

Exp. 20) A train crosses a man coming from the opposite direction in 7.5 seconds. If the speed of man be 10 m/s and speed of train is 20 m/s, find the length of the train.

Solution Length of train = Time × Relative speed
$= 7.5 \times (10 + 20) = 7.5 \times 30 = 225 \text{ m}$

Exp. 21) A train of length 250 m crosses a bridge of length 150 m in 20 seconds. What is the speed of train?

Solution (Length of train + Length of bridge)
= Speed of train × Time
$(250 + 150) = 20 \times \text{Speed}$
$\text{Speed} = \frac{400}{20} = 20 \text{ m/s} = 72 \text{ km/h}$

Exp. 22) Two trains coming from the opposite sides crosses each other in 10 seconds if the lengths of first train and second train be 125 m and 175 m respectively, also the speed of first train be 36 km/h, find the speed of second train.

Solution Speed of first train = 36 km/h = 10 m/s

Now, $\text{Time} = \frac{\text{Sum of length of the two trains}}{\text{Sum of their speeds}}$

$10 = \frac{125 + 175}{(10 + x)} \Rightarrow x = 20 \text{ m/s} = 72 \text{ km/h}$

Exp. 23) A fast moving superfast express crosses another passenger train in 20 seconds. The speed of faster train is 72 km/hr and speeds of slower train is 27 km/h. Also the length of faster train is 100 m, then find the length of the slower train if they are moving in the same direction.

Solution $\text{Time} = \frac{\text{Sum of length of the two train}}{\text{Difference in speeds}}$

$20 = \frac{(100 + x)}{25/2} \Rightarrow x = 150 \text{ m}$

NOTE Relative speed = (72 − 27) = 45 km/h
$= 45 \times \frac{5}{18} = \frac{25}{2} \text{ m/s}$

9.5 Concept Based on Boats and Rivers (or streams)

(i) When the boat and stream (or current) of river move in the same direction, then the relative speed of the boat is the sum of the individual speeds of boat and river. It is known as **downstream speed**.

(ii) When the boat moves against the current of the river (*i.e.,* in opposite direction), then the relative speed of the boat is the difference of the speeds of the boat and stream (of the river). It is known as **upstream speed**.

Let the speed of boat in still water be B and speed of current of river be C then,

$$\begin{cases} \text{Downstream speed} = (B + C) \\ \text{Upstream speed} = (B - C) \end{cases} ; B > C$$

$$\begin{cases} \text{Speed of the boat in still water} = \frac{(D + U)}{2} \\ \text{Speed of current (or stream)} = \frac{(D - U)}{2} \end{cases}$$

where $D \rightarrow$ downstream speed of the boat

and $U \rightarrow$ upstream speed of the boat

When the distance covered by boat in downstream (*i.e.,* with the flow of water) is same as the distance covered by boat in upstream (against the flow of the water) then,

$$\frac{\text{Time taken by boat in DS}}{\text{Time taken by boat in US}} = \frac{\text{Upstream speed}}{\text{Downstream speed}}$$

DS $\rightarrow$ Downstream, US $\rightarrow$ Upstream

Exp. 24) A boat can move at 5 km/h in still water (i.e., when water is not flowing). The speed of stream of the river is 1 km/h. A boat takes 80 minutes to go from a point A to another point B and return to the same point.

(i) What is the distance between the two points?
(ii) What is the ratio of downstream speed and upstream speed?
(iii) What is the ratio of time taken in downstream speed to the upstream speed?

Solution Downstream speed of boat = (5 + 1) = 6 km/h
Upstream speed of boat = (5 − 1) = 4 km/h

Therefore, $\frac{\text{Downstream speed}}{\text{Upstream speed}} = \frac{\text{Upstream time}}{\text{Downstream time}}$

$$\frac{6}{4} = \frac{3}{2} = \frac{\text{Time taken in upstream direction}}{\text{Time taken in downstream direction}}$$

$\therefore$ Time taken in downstream $= \frac{2}{5} \times 80 = 32 \text{ min}$

and time taken in upstream direction $= \frac{3}{5} \times 80 = 48 \text{ min}$

$\therefore$ Distance between two points = DS speed × DS time

$$= \text{US speed} \times \text{US time}$$

where DS → Downstream and US → Upstream

$$D = \frac{6 \times 32}{60} = 3.2\,\text{km}$$

or $$D = 4 \times \frac{48}{60} = 3.2\,\text{km}$$

(i) 3.2 km (ii) 3 : 2 (iii) 2 : 3

Exp. 25) A man can row 9 km/h in still water. It takes him twice as long as to row up as to row down. Find the rate of stream of the river.

Solution $\frac{\text{Time taken in upstream}}{\text{Time taken in downstream}} = \frac{2}{1}$

$\therefore$ $\frac{\text{Downstream speed}}{\text{Upstream speed}} = \frac{2}{1}$ where $\frac{B+R}{B-R} = \frac{2}{1}$

$B \rightarrow$ Speed of boat in still water

$R \rightarrow$ Speed of current

$\Rightarrow$ $\frac{B}{R} = \frac{3}{1}$ (By componendo and dividendo)

$\Rightarrow$ $\frac{9}{R} = \frac{3}{1} \Rightarrow R = 3$ km/h

9.6 Races

Terminology

(i) **Startup or headstart** When a runner allows to another runner to stay ahead in the same race, then it is said that there is a startup in the race.

For example if A allows B to go ahead before starting the race, then it is said that A gives startup to B and B has the startup. If before starting the race B goes ahead of x metre, then we can say A gives x metre startup to B or B has startup (or headstart) of x metre.

(ii) **Dead heat** When the runners reach the finishing line (or the final post) then it is said that these runners finish (or end) the race in dead heat.

Some More Useful Concepts

(i) When it is said that A can give B a start of x metre in y metre race, then it means in y metre race B runs x metre less than A in the same time.

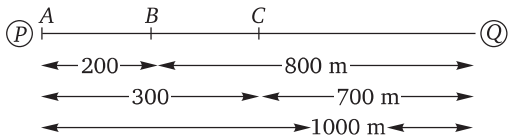
NOTE There is a great difference between 'can' and 'gives'.

(ii) When A beats B by t seconds in a race of y metre then it means B is the loser and A is the winner and when A reaches the finishing line B is still some distance back to A. Thus B takes t seconds to cover the remaining distance. Hence, we can calculate the speed of loser (B).

(iii) Throughout the race there is always a certain relationship among runners *i.e.*, they always maintain the ratio of speeds. I think this is the nub of all the problems pertaining to the races. Go ahead…

Exp. 1) A can give B a 200 m startup and C a 300 m startup in a race of 1 km. How many metres startup can B gives to C in a 1 km race.

Solution



Ratio of speeds of $A : B = 1000 : 800 = 5 : 4$

Ratio of speeds of $A : C = 1000 : 700 = 10 : 7$

Ratio of speeds of $B : C = 800 : 700 = 8 : 7$

Since, when B moves 8 m, C moves 7 metre. Therefore, when B moves 1000 m, C moves 875 metre. Thus, B can give C a start of 1000 − 875 = 125 m.

Alternatively Since, C is 12.5% slower than B. So, C will cover 12.5% less distance than B in the same time.

(Since when time is constant, they cover the distances in the ratio of their speeds.)

Thus, in 1000 m (or 1 km) when B runs 1000 m, C will run 125 m less than B. Hence, B can give a start of 125 m to C in a 1 km race.

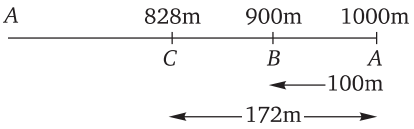
Exp. 2) In a one km race A gives B a start of 100 m and in a one km race B gives a start of 80 m to C. In a 1 km race who will win and by how much distance from the worst performer between two losers?

Solution Ratio of speeds of $A : B = 1000 : 900 = 100 : 90$

Ratio of speeds of $B : C = 1000 : 920 = 100 : 92$

Therefore, when A moves 1000 m, B moves 900 m and when B moves 900 m, C moves 828 m.

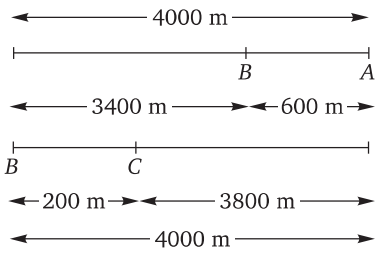
Thus,



Since, C moves 8% less than B in the same time. Thus, C is the worst performer and A will win by him by 172 m.

Exp. 3) In a 4 km race A wins by 600 m over B. B can give start of 200 m to C in a 4 km race. By how much distance C gets start up so that the race between A and C ends in dead heat in the same race of 4 km?

Solution



Ratio of speeds of $A : B = 20 : 17$

and Ratio of speeds of $B : C = 20 : 19$

$\therefore$ Ratio of speeds of $A : B : C = 400 : 340 : 323$

Therefore, in 4000 m race A run 4000 m, B run 3400 m and C run 3230 m. Thus C can get 770 m start up from A.

Exp. 4) In a 1500 m race A wins over B by 350 m and in 1500 m race C can give a startup of 250 m to B. By how much distance A give start up to C, so that A beats C by 50 metres?

Solution Ratio of speeds of $A : B = 30 : 23$

Ratio of speeds of $B : C = 5 : 6$

$\therefore$ Ratio of speeds of $A : B : C = 150 : 115 : 138$

So, when A moves 1500 m, B moves 1150 m and C moves 1380 m. Thus C moves 120 m less than A. To win A just by 50 m over C, A should give $120 - 50 = 70$ m startup to C.

Exp. 5) In a race of 2500 m, A beats B by 500 m and in a race of 2000 m, B beats C by 800 m. By what distance A gives startup to C so that they will end up with dead heat in 3 km race. Also find that by what distance A will win over C in a 1 km race?

Solution Ratio of speeds of $A : B = 5 : 4$

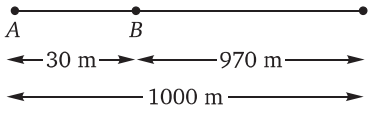
Ratio of speeds of $B : C = 5 : 3$

$\therefore$ Ratio of speeds of $A : B : C = 25 : 20 : 12$

In 3 km race A run 3000 m, B run 2400 m, C run 1440 m, so to end up the race in dead heat A should give C the startup of 1560 m and therefore in 1 km the same will be 520 m.

Exp. 6) A gives B, a start of 30 m or 10 seconds and end up the race of 1 km in dead heat. What is the ratio of speeds of A and B?

Solution



Since, either B has the startup of 30 m or 10 seconds. It means B runs 30 m in 10 seconds. Hence, the speed of B is 3 m/s.

NOTE Don't be confused that A's speed = 3 m/s.

Also, it is very simple when A moves 1000 m, B moves 970 m. Since, the ratio of speeds is equal to the ratio of distances covered by A and B in the same time. Thus, the speed of

$$A : B = 1000 : 970 = 100 : 97$$

Exp. 7) In a 1 km race A wins over B by 80 m or 20 seconds. B can give a start of 100 m to C in 1 km race. Find out that by how much time A will win over C? Also, find the ratio of speeds of B and C.

Solution Ratio of speeds of $A : B = 100 : 92$

Ratio of speeds of $B : C = 10 : 9$

$\therefore$ Ratio of speeds of $A : B : C = 1000 : 920 : 828$

Also, $$\text{Speed of } B = \frac{80}{20} = 4 \text{ m/s}$$

Therefore, Speed of $C = 3.6$ m/s

Now, C has to cover 172 m distance in extra time. So, the time taken by C to cover the remaining distance

$$= \frac{172}{3.6} = 47.77 \text{ s.}$$

Ratio of speeds of $B : C$ = Ratio of distances covered by $B : C$

$$= 1000 : 900 = 10 : 9$$

$\therefore$ $$\frac{\text{Speed of } B}{\text{Speed of } C} = \frac{10}{9}$$

Exp. 8) A can win B by 250 m in a 2 km race. What should be the change in distance of startup? So, that B must cover 20% less distance than that by A in the same time.

Solution $$\frac{\text{Distance covered by } A}{\text{Distance covered by } B} = \frac{5}{4}$$

$\therefore$ $$\frac{\text{Speed of } A}{\text{Speed of } B} = \frac{5}{4}$$

So, when A moves 2000 m, B should move 1600 m. But since initially B moves 1750 m. Therefore the new startup will be increased by 150 m.

Exp. 9) The ratio of speeds of A and B is 4 : 7 and A loses the race by 270 m, then what is the length of the race course?

Solution When B moves 7 m, A moves only 4 m. Hence, A loses the race by 3 m.

Now, since B loses by 3 m in the race of 7 m.

$\therefore$ B will lose 270 m in the race of 630 m.

Exp. 10) The ratio of time taken to run a certain distance by Pythagorus and Hawkins is 4 : 3 and thus Hawkins wins the race by 360 m. What is the distance of race course?

Solution $$\frac{\text{Time taken by Pythagorus}}{\text{Time taken by Hawkins}} = \frac{4}{3}$$

$$\frac{\text{Speed of Pythagorus}}{\text{Speed of Hawkins}} = \frac{3}{4}$$

$$= \frac{\text{Distance covered by P}}{\text{Distance covered by H}}$$

Now, when Hawkins runs 4 m, Pythagorus runs 3 m and thus Hawkins wins by 1 m.

So, when Hawkins wins the race by 1 m, race course is 4 m.

when Hawkins wins the race by 360 m, race course is

$$360 \times 4 = 1440 \text{ m}$$

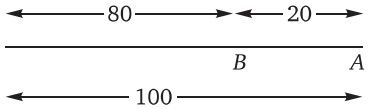
Alternatively 360 = 25% of the total length of race.

$\therefore$ length of race course = 1440 m.

Exp. 11) In a race of D km, A wins over B by 0.2 D and in the same length of race B wins over C by 0.25 D. What should be the head- start to C, So that A and C finish the race at the same time.

Solution **NOTE** Instead of solving the problem using 'D' as a distance convert it into 100 *i.e.*, suppose $D = 100$.
(It is just for your convenience.)

Now,



Now, Speed of $A : B = 100 : 80 = 5 : 4$

Speed of $B : C = 4 : 3$

$\therefore$ Speed of $A : B : C = 20 : 16 : 12$

$= 100 : 80 : 60$

$\{40\}$

So, A can give 'C' a start of 40 m in 100 m race. Therefore, A can give 0.4 D. Start to 'C' in the race of D unit.

Exp. 12) In a 3 km race the speeds of A and B are in the ratio of 6 : 5 and A wins by 10 seconds. What is the time taken by B to finish the race. Also, to end the race in dead heat what per cent of total distance A should give the startup to B?

Solution Ratio of distances covered by A and B when A just reaches the finishing line = 6 : 5.

Thus, B has to cover 500 m distance in extra time and this 500 m distance is covered by B in 10 seconds.

So, the speed of $B = 50$ m/s.

Thus, the total time required by B to complete the 3 km race

$$= \frac{3000}{50} = 60\text{ s} = 1\text{ min}$$

Since, B coveres 16.66% distance less than A covers in the same time, so A should give 16.66% of the total distance as a startup to B.

Exp. 13) In a 1 km race A gives B a startup of 5 seconds and still wins over B by 15 seconds. The ratio of speeds of A and B is 2 : 1. Find the time taken by A to finish 2.5 km race.

Solution

$$\frac{\text{Speed of } A}{\text{Speed of } B} = \frac{2}{1}$$

$$\frac{\text{Time taken by } A}{\text{Time taken by } B} = \frac{1}{2}$$

$$\frac{t}{(t+20)} = \frac{1}{2}$$

$$\Rightarrow \quad t = 20\text{ s}$$

Thus, A needs 20 seconds to cover 1 km. Thus, to cover 2.5 km race he needs $20 \times 2.5 = 50$ seconds.

Exp. 14) X can beat Y by 200 m in a race of 2000 m. Y can beat Z by 100 m in a race of 2500 m. By how many metres can X beat Z in a race of 1000 m.

Solution Ratio of speeds of X and $Y = 10 : 9$ (2000 : 1800)

Ratio of speeds of Y and $Z = 25 : 24$ (2500 : 2400)

Ratio of speeds of X, Y and $Z = 250 : 225 : 216$

Since in a race of 250 m, X beats Z by 34 m.

So, in a race of 1000 m, X will beat Z by 136 m.

Exp. 15) In a 6 km race B has 250 m headstart and C has 500 m headstart by A, still A beats C and B by 235 m and 350 m respectively. How many metres startup can B give to C so as to end up the race at the same time with C in the race of 6 km. Also find the ratio of speeds of $A : B : C$.

Solution When A runs 6000 m, B runs 5400 m only and C runs 5265 m only.

So, when B runs 6000 m, C will run 5850 m. So, B can give 150 m startup to C.

Ratio of speeds of $A : B : C = 6000 : 5400 : 5265$

$= 1200 : 1080 : 1053$

$= 400 : 360 : 351$

Exp. 16) A can run 1 km in 2 min 20 second and B can run the same distance in 3 min. What is the distance travelled by B in the same time as A travels, when they start simultaneously in the race of 4.5 km.

Solution

$$\frac{\text{Time taken by } A}{\text{Time taken by } B} = \frac{140s}{180s} = \frac{7}{9}$$

$$\therefore \quad \frac{\text{Speed of } A}{\text{Speed of } B} = \frac{\text{Distance travelled by } A}{\text{Distance travelled by } B} = \frac{9}{7}$$

Therefore, B travels $\frac{7}{9} \times 4.5 = 3.5$ km.

Exp. 17) Shahrukh takes 4 min to cover the same distance for which Urmila takes 6 min 30 sec. What is the ratio of distances covered by Shahrukh and Urmila in the race of 2.6 km and by what distance Shahrukh wins over Urmila?

Solution

$$\frac{\text{Time taken by Shahrukh}}{\text{Time taken by Urmila}} = \frac{240}{390} = \frac{8}{13}$$

$$\frac{\text{Distance covered by Shahrukh}}{\text{Distance covered by Urmila}} = \frac{13}{8}$$

Hence, Shahrukh will win the race by 1 km.

Exp. 18) A runs at the speed of $\frac{7}{4}$ times the speed of B. By calculation B finds that she has to run 300 m after A reaches to the winning post. What is the total distance of race?

Solution

$$\frac{\text{Speed of } A}{\text{Speed of } B} = \frac{7}{4}; \frac{\text{Distance of } A}{\text{Distance of } B} = \frac{7}{4}$$

So, the total distance of race = 700 m.

$(\because\ 7x - 4x = 300 \Rightarrow 7x = 700)$

Exp. 19) Time taken by A is 5/7 of B's time for the same length of race. The speed of A is 84 m/s and A beats B by 240 m. What is the length of race course?

Solution

$$\frac{\text{Speed of } A}{\text{Speed of } B} = \frac{\text{Time taken by } B}{\text{Time taken by } A} = \frac{7}{5}$$

Now since, $7x - 5x = 240$

$\Rightarrow \quad 7x = 840$ m

Exp. 20) In a race of 3000 m, Michal beats Nicholas by 600 m and Nicholas in a race of 2000 m beats Oscar by 600 m. In a 1 km race by how much distance Michal beats Oscar?

Solution Ratio of speeds of $M : N : O = 100 : 80 : 56$

So, Michal (M) beats Oscar (O) by 440 m in 1 km race.

Exp. 21) In a race of 500 m, President runs at 5 m/s. If president gives chairman a start of 20 m and still beats him by 20 seconds, what is the chairman's speed?

Solution Chairman has to cover only 480 m. He takes total

$$100 + 20 = 120 \text{ s}$$

Therefore, speed of chairman $= \frac{480}{120} = 4 \text{ m/s}$

Exp. 22) In a race of 800 m Dholakiya gives Preetam a start of 200 m and then loses the race by 20 seconds. What is the speed of Preetam and Dholakiya respectively? If the ratio of respective speeds be 3 : 2.

Solution Let the speed of Dholakiya be S_D and speed of Preetam be S_P and let the time taken by Preetam be t second

then, $$\frac{S_D}{S_P} = \frac{\frac{800}{(t+20)}}{\frac{600}{t}} = \frac{2}{3}$$

$$\frac{800}{600} \times \frac{t}{(t+20)} = \frac{2}{3}; t = 20 \text{ s}$$

$\therefore$ $$\text{Speed of Preetam} = \frac{600}{20} = 30 \text{ m/s}$$

and $$\text{speed of Dholakiya} = \frac{800}{40} = 20 \text{ m/s}$$

Exp. 23) In race of 1 km Sahara gives Birla a startup of 250 m still Sahara wins by 150 m. What is the ratio of speeds of Sahara and Birla?

Solution When Sahara covers 1000 m, Birla covers only 600 m.

So, the ratio of speeds of Sahara and Birla = 1000 : 600 = 5 : 3

Exp. 24) Priyambda wins the race over Kokilaben by 150 m in a race of 1 km but when she gives a startup of 5 seconds to Kokilaben she wins by 65 m. Find the speed of Kokilaben.

Solution In 5 seconds Kokilaben runs 85 m.

So, the speed of Kokilaben = 17 m/s.

9.7 Circular Motion

(i) When the bodies move in the opposite direction, their relative speed becomes equal to the sum of their individual speed.

(ii) When two bodies move in the same direction their relative speed becomes equal to the difference of the individual speeds.

First Meeting

(i) Let A and B be two runners. Time taken by them to meet for the first time $= \frac{\text{Length of the circular track}}{\text{Relative speed}}$

(ii) When there are more than two runners, then suppose A is the fastest runner and A meets B for the first time in t_{AB} hours and A meets C for the first time in t_{AC} hours and A meets D for the first time in t_{AD} seconds/hours and so on. Then time taken by all of them to meet for the first time is the LCM of t_{AB}, t_{AC}, t_{AD} etc.

First Meeting at the Starting Point

Let A takes t_A time to complete one round and B takes t_B time and C takes t_C time and so on, then the time taken to meet for the first time at the starting point = LCM of t_A, t_B, t_C etc.

Exp. 1) Arjun and Bhishma are running on a circular track of length 600 m (i.e., circumference of the track is 600 m). Speed of Arjun is 75 m/s and that of Bhishma is 45 m/s. They start running from the same point at the same time the same direction.

(i) When will they meet again for the first time?

(ii) When will they meet again for the second time?

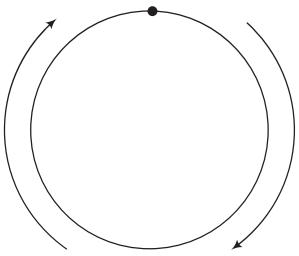
(iii) When will they meet again for the tenth time?

Solution Since, the two bodies are continuously and consistently running on the same circular path they will certainly meet again and again.

Under the given circumstances the two bodies will meet, after the start, when the difference in distance traversed by them is either 1 round or 2 rounds or 3 rounds and so on.

That is, their first meeting occurs when the faster body covers 1 extra round. And, second meeting occurs when the faster body covers 2 extra rounds. And, third meeting occurs when the faster body covers 3 extra rounds.

And so on so forth. In other words, the faster body has to have a lead of 1 round, 2 rounds, 3 rounds etc. depending on the number/order of meeting. And, thus the time required to have a lead of 1 round $= \frac{\text{Circumference}}{\text{Relative speed}}$. So, the time required to have a lead of n rounds $= n \times \frac{\text{Circumference}}{\text{Relative speed}}$



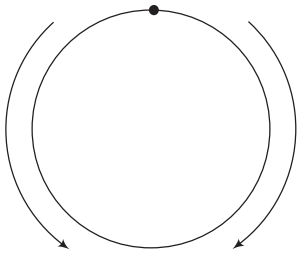
(i) In order to meet for the first time, after the start of the race, Arjun (faster man) has to have a lead of 600 m, because as soon as Arjun will cover 600 m extra than that of Bhishma, Arjun will overtake Bhishma.
And, you know that Arjun gets a lead of 30 m in every 1 second, so to have a lead of 600 m Arjun has to run for 600/30 = 20 seconds.
Otherwise, $\text{Time} = \frac{\text{Circumference}}{\text{Relative speed}} = \frac{600}{75-45} = \frac{600}{30} = 20$ seconds
That is after 20 seconds they will meet for the first time, while running in the same direction.

(ii) In order to meet for the second time, after the start of the race, Arjun (faster man) has to have a lead of 2 rounds.
$$\text{Time} = 2 \times \frac{\text{Circumference}}{\text{Relative speed}} = 2 \times \frac{600}{75-45} = \frac{1200}{30} = 60 \text{ seconds}$$
That is after 60 seconds they will meet for the second time, while running in the same direction.

(iii) In order to meet for the tenth time, after the start of the race, Arjun (faster man) has to have a lead of 10 rounds.
$\text{Time} = 10 \times \frac{\text{Circumference}}{\text{Relative speed}} = 10 \times \frac{60}{75-45} = \frac{6000}{30} = 200$ seconds
That is after 200 seconds they will meet for the tenth time, while running in the same direction.

Exp. 2) Arjun and Bhishma are running on a circular track of length 600 m. Speed of Arjun is 75 m/s and that of Bhishma is 45 m/s. They start running from the same point at the same time in the opposite directions.
(i) When will they meet again for the first time?
(ii) When will they meet again for the second time?
(iii) When will they meet again for the tenth time?

Solution Since two bodies are continuously and consistently running on the same circular path they will certainly meet again and again.
Under the given circumstances the two bodies will meet, after the start, when the total distance traversed by them is either 1 round or 2 rounds or 3 rounds and so on.
That is their first meeting occurs when they together cover 1 round.
And, second meeting occurs when they together cover 2 rounds.
And, third meeting occurs when they together cover 3 rounds.
And so on so forth.
In other words, they have to complete together 1 round, 2 rounds, 3 rounds etc. depending on the number/order of meeting.
And, thus the time required to complete 1 round
$$= \frac{\text{Circumference}}{\text{Relative speed}}$$
So, the time required to complete n rounds
$$= n \times \frac{\text{Circumference}}{\text{Relative speed}}$$



(i) In order to meet for the first time, after the start of the race, Arjun and Bhishma together will have to traverse together 600 m.
And, you know that they together traverse 120 m in every 1 second, so to traverse 600 m they have to run for 600/120 = 5 seconds.
Otherwise, $\text{Time} = \frac{\text{Circumference}}{\text{Relative speed}} = \frac{600}{75+45} = \frac{600}{120} = 5$ seconds
That is after 5 seconds they will meet for the first time, while running in the opposite directions.

(ii) In order to meet for the second time, after the start of the race, Arjun and Bhishma together will have to traverse 2 times a round.
$$\text{Time} = 2 \times \frac{\text{Circumference}}{\text{Relative speed}} = 2 \times \frac{600}{75+45} = \frac{1200}{120} = 10 \text{ seconds}$$
That is after 10 seconds they will meet for the second time, while running in the opposite directions.

(iii) In order to meet for the tenth time, after the start of the race, Arjun and Bhishma together will have to traverse 10 times a round.
$$\text{Time} = 10 \times \frac{\text{Circumference}}{\text{Relative speed}} = 10 \times \frac{60}{75+45} = \frac{6000}{120} = 50 \text{ seconds}$$
That is after 50 seconds they will meet for the tenth time, while running in the opposite directions.

Exp. 3) Arjun and Bhishma are running on a circular track of length 600 m. Speed of Arjun is 75 m/s and that of Bhishma is 45 m/s. They start running from the same point at the same time.
(i) When will they meet again for the first time at the starting point?
(ii) When will they meet again for the second time at the starting point?
(iii) When will they meet again for the tenth time at the starting point?

Solution Since, we are not interested in finding the meetings that may occur somewhere else on the track other than the starting point, so it does not matter whether they are moving in the same direction or in the opposite directions.
It implies that we need to know when the two bodies are completing the rounds together, though it is not necessary and possible that the numbers of rounds to be same, as their speeds are distinct.
So, we have to find out when the two bodies complete their rounds and then we can find out the common time when both of them will complete their rounds.
So you can see that Arjun completes a round in every 8 (= 600/75) seconds.
And, Bhishma completes a round in every 40/3 (= 600/45) seconds.

	Time taken to Complete the Different Rounds $(R_1, R_2, \ldots)$									
	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}
Arjun	8	16	24	32	40	48	56	64	72	80
Bhishma	$\frac{40}{3}$	$\frac{80}{3}$	40	$\frac{160}{3}$	$\frac{200}{3}$	80	$\frac{280}{3}$	$\frac{320}{3}$	120	$\frac{400}{3}$

Thus, it is obvious that the two bodies meet at the starting point every 40 seconds, which is nothing but the LCM of 8 seconds and 40/3 seconds.
Otherwise, Time taken by Arjun to complete 1 round

$$= \frac{600}{75} = 8 \text{ seconds}$$

And, time taken by Bhishma to complete 1 round

$$= \frac{600}{45} = \frac{40}{3} \text{ seconds}$$

LCM of 8 and 40/3 = 40

(i) Therefore, the time required for the first meeting at the starting point = 40 seconds
(ii) Therefore, the time required for the second meeting at the starting point = 2×40 = 80 seconds
(iii) Therefore, the time required for the tenth meeting at the starting point = 10×40 = 400 seconds

Exp. 4) Arjun, Bhishma and Nakul run on the circular path at the speed of 20 m/s, 30 m/s and 50 m/s respectivley in the same direction. The circumference of the track (or path) is 600 m.

(i) When will they be together again for the first time?
(ii) When will they be together again for the first time at the starting point?

Solution

(i) Nakul meets Arjun after every $= \frac{600}{(50-20)} = 20$ s

Nakul meets Bhishma after every $= \frac{600}{(50-30)} = 30$ s

Therefore, all of the three would meet after every 60 seconds. (60 = LCM of 20 and 30). Hence, they would all meet for the first time after 60 seconds.

(ii) Arjun takes 600 / 20 = 30 s to complete one round

Bhisma takes $\frac{600}{30} = 20$ s to complete one round and

Nakul takes $\frac{600}{50} = 12$ s to complete one round

Hence, they would meet for the first time at the starting point after 60 seconds.

Hint 60 = LCM of 30, 20 and 12.

Exp. 5) Arjun, Bhishma, Chaitanya and Duryodhana are running on a circular track of length 600 m. They start running from the same point at the same time in the same direction and the Speeds of Arjun, Bhishma, Chaitanya and Duryodhana are 45 m/s, 55 m/s, 65 m/s and 75 m/s, respectively.

(i) When will they meet again for the first time?
(ii) When will they meet again for the seventh time?
(iii) When will they meet again for the first time at the starting point?
(iv) When will they meet again for the seventh time at the starting point?

Solution Let us, first of all, find the answers for (i) and (ii) parts.
Since, Duryodhana is the fastest runner, so we will find out the time taken by Duryodhana to meet Arjun, Bhishma and Chaitanya independently, then we will find out the common time when Duryodhana meets all the three together.
Time taken by Duryodhana to meet Arjun for the first time

$$= \frac{600}{75-45} = \frac{600}{30} = 20 \text{ seconds}$$

Time taken by Duryodhana to meet Bhishma for the first time

$$= \frac{600}{75-55} = \frac{600}{20} = 30 \text{ seconds}$$

Time taken by Duryodhana to meet Chaitanya for the first time

$$= \frac{600}{75-65} = \frac{600}{10} = 60 \text{ seconds}$$

Now, the LCM of 20, 30 and 60 is 60, so Duryodhana will meet all of them together in every 60 seconds.

(i) It can be conclude that all of them meet for the first time after 60 seconds.
(ii) It can be conclude that all of them meet for the seventh time after 420 (= 7×60) seconds.

Now, let us solve it for the (iii) and (iv) part.
Since, we are interested in finding the meetings occurring at the starting point, so the direction of their motion does not matter. Therefore, we will find out the time taken by each of them to complete a round and then we will find out when all of them will complete the rounds together, though it is not necessary and possible that the number of rounds to be same, as their speeds are distinct.

Time taken by Arjun to complete 1 round $= \frac{600}{45} = \frac{40}{3}$ seconds

And, time taken by Bhishma to complete 1 round

$$= \frac{600}{55} = \frac{120}{11} \text{ seconds}$$

Time taken by Chaitanya to complete 1 round

$$= \frac{600}{65} = \frac{120}{13} \text{ seconds}$$

And, time taken by Duryodhana to complete 1 round

$$= \frac{600}{75} = 8 \text{ seconds}$$

Now, the LCM of 40/3, 120/11, 120/13 and 8 = 120

Thus, all of them will meet at the starting point in every 120 seconds.

(iii) It can be concluded that all of them meet for the first time after 120 seconds.

(iv) It can be concluded that all of them meet for the first time after 840 (= 7 × 120) seconds

Exp. 6) Arjun, Bhishma, Chaitanya, Duryodhana and Eklavya are running on a circular track of length 600 m. They have started running from the same point at the same time and the Speeds of Arjun, Bhishma, Chaitanya, Duryodhana and Eklavya are 25 m/s, 35 m/s, 45 m/s, 60 m/s and 75 m/s, respectively. Arjun and Chaitanya are running clockwise, while Bhishma, Duryodhana and Eklavya are running anti-clockwise.

(i) When will they meet again for the first time?

(ii) When will they meet again for the fifth time?

Solution Since Eklavya is the fastest runner, so we will find out the time taken by Eklavya to meet Arjun, Bhishma, Chaitanya and Duryodhana independently, then we will find out the common time when Eklavya meets all of them together.

Time taken by Eklavya to meet Arjun for the first time

$$= \frac{600}{75 + 25} = \frac{600}{100} = 6 \text{ seconds}$$

Time taken by Eklavya to meet Bhishma for the first time

$$= \frac{600}{75 - 35} = \frac{600}{40} = 15 \text{ seconds}$$

Time taken by Eklavya to meet Chaitanya for the first time

$$= \frac{600}{75 + 45} = \frac{600}{120} = 5 \text{ seconds}$$

Time taken by Eklavya to meet Duryodhana for the first time

$$= \frac{600}{75 - 60} = \frac{600}{15} = 40 \text{ seconds}$$

Now, the LCM of 6, 15, 5 and 40 is 120, so Duryodhana will meet all of them together in every 120 seconds.

(i) It can be conclude that all of them meet for the first time after 120 seconds.

(ii) It can be conclude that all of them meet for the fifth time after 600 (= 5 × 120) seconds.

Exp. 7) Two girls A and B run on a circular track, with constant speeds, such that the ratio of their speeds is 3:5.

(i) Find the maximum number of distinct points where they can meet, if they run in the same direction.

(ii) Find the maximum number of distinct points where they can meet, if they run in the opposite directions.

Solution When two bodies move in the same direction, the maximum number of distinct meeting points = difference in the ratio = 5 − 3 = 2. When two bodies move in the opposite directions, the maximum number of distinct meeting points = sum of the ratio = 5 + 3 = 8

Exp. 8) Two girls A and B run on a circular track, with the speeds of 80 m/s and 140 m/s.

(i) Find the maximum number of distinct points where they can meet, if they run in the same direction.

(ii) Find the maximum number of distinct points where they can meet, if they run in the opposite directions.

Solution When two bodies move in the same direction, the maximum number of distinct meeting points = difference in the ratio = 7 − 4 = 3. When two bodies move in the opposite directions, the maximum number of distinct meeting points = sum of the ratio = 7 + 4 = 11

Exp. 9) Two hands of a normal clock run at the speed of 30° per hour and 360° per hour. Find the number of distinct points where they meet on the dial.

Solution When two bodies move in the same direction, the maximum number of distinct meeting points = difference in the ratio = 12 − 1 = 11. It means, in a normal clock minute hand meets hour hand at 11 distinct points on the dial.

Exp. 10) There are two planets *A* and *B* revolving around a star in the two concentric orbits at the same plane. Planet *A* takes 366 days and *B* takes 30 days to complete a revolution around that star. But, whenever they pass each other the people living on the planet *A* observe the eclipse. Consider that a year has 366 days.

(i) Find the number of times when planet *A* experiences the eclipse in a year, provided both the planets move in the same direction.

(ii) Find the number of times when planet *A* experiences the eclipse in a year, provided the two planets move in the opposite directions.

Solution When two bodies move in the same direction, the maximum number of distinct meeting points = difference in the ratio = 61 − 5 = 56. Therefore, during a year planet *A* experiences 56 eclipses. When two bodies move in the opposite directions, the maximum number of distinct meeting points = sum of the ratio = 61 + 5 = 66

Therefore, during a year planet *A* experiences 66 eclipses.

Exp. 11) A tortoise can complete a full round on the circular track in 8 min, while the rabbit can do the same on the same circular track in 5 min. *A*, *B*, *C* and *D* are the four consecutive points on the circular track which are equidistant from each other. *A* is opposite to *C* and *B* is opposite to *D*.

(i) After how many minutes will they meet together for the first time, when both have started simultaneously from the same point in the same direction?

(ii) After how many minutes will they meet together for the first time at the starting point, when both have started simultaneously from the same point in the same direction?

(iii) If they start from a point A, simultaneously in the same direction then after how many minutes will they meet at C which is just opposite to A?

(iv) If tortoise has a lead of 5 min, when would they meet for the first time, where they have started from the same point in the same direction?

Solution

(i) Ratio of time of rabbit and tortoise = 5 : 8

∴Ratio of the speed of rabbit and tortoise = 8 : 5

Let the length of the circular track be 40 m, then the speed of rabbit = 8 m/min and speed of tortoise = 5 m/min. (40 is the LCM of 8 and 5, taken just for conveninence in calculation)

Now, the time when they will be together for the first time

$$= \frac{\text{Circumference}}{\text{Relative speed}} = \frac{40}{3}\text{ min}$$

(ii) Time taken by rabbit to complete one round = 5 min

Time taken by tortoise to complete one round = 8 min

∴ They will meet at the starting point, for the first time

= LCM of 5 and 8 = 40 min

(iii) Rabbit reaches point *C* in $\frac{20}{8} = 2.5$ min.

Rabbit reaches the same point *C* in 7.5, 12.5, 17.5, 22.5 min etc. Tortoise reaches after 4 min, 12 min, 20 min, … etc.

They will never meet at *C* since time cannot coincide as time taken by rabbit is the decimal number and time taken by tortoise is a natural number.

(iv) In 5 min tortoise travels 25 m. Now, rabbit starts chasing tortoise @ 8 m/min.

∴ Time taken by rabbit to overtake tortoise $= \frac{25}{(8-5)} = \frac{25}{3}$ min

Exp. 12) Alfred and Bernard run on the circular track of 600 m. Speeds of Alfred and Bernard are 30 m/s and 20 m/s respectively. Initially they are diametrically opposite to each other.

(i) When will they meet for the first time if both move in the same direction?

(ii) If both of them move in opposite directions, when will they meet for the second time?

Solution

(i) Relative speed of Alfred = 30 – 20 = 10 m/s

Bernard is 300 m ahead in the race.

So, time taken by Alfred to catch Bernard = 300/10= 30 s

(ii) To meet for the second time they have to cover

= 300 + 600 = 900 m

there relative speed = 30 + 20 = 50 m/s

(since direction is opposite)

∴ Time taken to meet for the second time $= \frac{900}{50} = 18$ s

Exp. 13) A, B and C run on a circular track of 800 m. Speeds of A, B and C are 20 m/s, 26 m/s and 33 m/s.

(i) When will they meet for the first time?

(ii) What is the ratio of distances covered by each one to meet for the first time.

Solution

(i) Time taken by *C* to meet *A* for the first time $= \frac{800}{33-20} = \frac{800}{13}$ s

Time taken by *C* to meet *B* for the first time $= \frac{800}{33-26} = \frac{800}{7}$ s

LCM of $\frac{800}{13}$ and $\frac{800}{7} = 800$ s.

Therefore, all of them (*i.e.*, *A*, *B* and *C*) meet after 800 seconds = 13 min 20 sec.

(ii) The ratio of distances = ratio of speeds = 20 : 26 : 33

Exp. 14) Two friends – Choar and Poohlice start walking simultaneously from a point on the circular track. Their friend Pubhlik observes from a distance that when they walk in the same direction they take 7 times the time they take when they walk in the opposite direction for their first meeting to happen. If the speed of Choar, who walks faster, is 16 m/s, what is the speed of Poohlice?

(a) 8 m/s (b) 10 m/s (c) 12 m/s (d) 14 m/s

Solution When they move in the same direction their effective speed is equal to the difference of their individual speeds and when they move in the same direction their effective speed is equal to the sum of their individual speeds. Also, speed is always inversely proportional to the time.

Now, let us assume that the speed of Poohlice be *p*, then

$$\frac{16+p}{16-p} = \frac{7}{1} \Rightarrow p = 12\text{ m/s}$$

Hence, choice (c) is the correct one.

Hint Time taken by them to meet for the first time

$$= \frac{\text{Length of the circular track}}{\text{Relative Speed}}$$

And since length of the circular track is constant irrespective of the direction, so the ratio of time duration to meet for the first time would depend upon the two distinct relative speeds.

$$\frac{\left(\frac{\text{Length of the circular track}}{16-p}\right)}{\left(\frac{\text{Length of the circular track}}{16+p}\right)} = \frac{7}{1}$$

$$\Rightarrow \quad \frac{16+p}{16-p} = \frac{7}{1}$$

Introductory Exercise 9.1

1. *P* and *Q* start running on the circular track in opposite directions. They start from a point *A* and meet for the first, second and third time at *B*, *C* and *A*, respectively. What is the ratio of speeds of *P* and *Q*?
   (a) 2 : 1 (b) 3 : 1
   (c) 4 : 1 (d) none of these
2. *P* and *Q* run on a circular track of 100 m. Speed of *P* is 250 m/s and that of *Q* is 400 m/s. At how many points on a circular track will they meet if they start from the same point, simultaneously, in the same direction?
   (a) 1 (b) 2
   (c) 3 (d) 4
3. *A*, *B* and *C* start walking around a circular track. *A* completes one round in 91/6 seconds, *B* completes in 65/4 seconds and *C* completes in 56/3 seconds respectively. After how many seconds will they meet each other at the starting point again?
   (a) 3460 s
   (b) 1820 s
   (c) 3640 s
   (d) none of the above

9.8 Clocks

Actually the movement of hour-hand and minute-hand follows the relative motion. The dial of the clock behaves like a circular track and where minute-hand is a faster runner and hour-hand is a slower one.

For better understanding with the clocks, assume 60 minutes shown on the dial as 60 points. Here we give an arbitrary new unit of distance as 'point'.

So, minute-hand (MH) runs on the circular track of 60 points at 60 points per hour and hour-hand (HH) runs at 5 points per hour. Now we become familiar with the relative motion of two hands of a clock.

Here, $1 \text{ point} = 6^\circ$ and $60 \text{ point} = 360^\circ$

also $1 \text{ point} = 6^\circ = 1 \text{ min}$

Exp. 1) How many times minute-hand coincides with hour-hand in 12 hours?

Solution Time taken by minute-hand to meet hour-hand for the first time $= \frac{60}{55} = \frac{12}{11}$ hours.

(Assume initially both hands are at 12 *i.e.*, 12 O′ clock is shown by them.)

Therefore, after every 12/11 hours-minute hand and hour-hand meet each other (or coincide or overtake).

Now, since in $\frac{12}{11}$ hours they coincide 1 time

So, in 12 hours they will coincide $= \frac{12}{12/11} = \frac{12}{12} \times 11 = 11$ times

Remember Between 11 O'clock and 1 O'clock, two hands coincide only one time, that's why they coincide with each other only 11 times in 12 hours.

NOTE In every $\frac{12}{11}$ hours (or $65\frac{5}{11}$ min) two hands of a clock coincide.

Exp. 2) In 12 hours how many times the two hands of clock will be just opposite to each other i.e., they make a straight line having the difference of 180° between them?

Solution $180^\circ = 30$ points

For the first time minute-hand and hour-hand will be separated in $\frac{30}{55}$ h $\left(\text{Time} = \frac{\text{Required distance}}{\text{Relative speed}}\right)$ and for every next time they will take $\frac{60}{55}$ minutes more to occur as opposite to each other. (Since in 60/55 hours they complete one round of clock's dial *i.e.*, 60 points undergoing the relative motion, we have total 12 hours.)

Thus, $\frac{30}{55} + \frac{60}{55} + \frac{60}{55} + \frac{60}{55} + \ldots = 12$

$\Rightarrow \quad \frac{30}{55} + \frac{60}{55} + \frac{60}{55} + \ldots = \frac{12 \times 55}{55}$

$\Rightarrow \quad 6 + 12 + 12 + \ldots = 12 \times 11 = 132$

$\Rightarrow \quad 6 + 12n = 132$

$\Rightarrow \quad 12n = 126 \quad \Rightarrow \quad n = 10$

only integral value is admissible.

Thus, total $10 + 1 = 11$ times both hands of a clock will be opposite to each other.

Remember: Between 5 O'clock and 7 O'clock the two hands make 180° angle only one time, that's why they make 180° angle only 11 times in 12 hours, *i.e.*, at exactly 6 O'clock they are 180° apart.

Exp. 3) In 12 hours how many times a minute-hand and hour-hand of a clock makes 90° between them or becomes perpendicular to each other?

Solution $90^\circ = 15$ points

This problem can be solved in two parts.

(i) When minute-hand goes ahead of hour-hand.

(ii) When hour-hand goes ahead of minute-hand.

(i) For the first time minute-hand and hour-hand will make 90° (or 15 points) difference in $\frac{15}{55}$ hours.

Since $\text{time} = \frac{\text{distance}}{\text{relative speed}}$

Now, after every $\frac{60}{55}$ hours they will occur at 90°. Since, in every $\frac{60}{55}$ hours they create a difference of 360° or 60 points (as the circumference of dial).

Now, we have 12 hours,

$$\therefore \quad \frac{15}{55}+\frac{60}{55}+\frac{60}{55}+\ldots = 12\,\text{h} = 12\times\frac{55}{55}$$

$$\Rightarrow \quad \frac{3}{11}+\frac{12}{11}+\frac{12}{11}+\ldots = \frac{12\times 11}{11}$$

$$\Rightarrow \quad \frac{3}{11}+\left(\frac{12}{11}n\right) = \frac{12\times 11}{11}$$

$$\Rightarrow \quad 3+12n = 132$$

$$\Rightarrow \quad n = \frac{129}{12} = 10$$

(only integral value of n is acceptable)

Therefore, $10 + 1 = 11$ times in 12 hours minute-hand makes 90° angle between the two hands, but when minute hand is ahead of hour-hand.

(ii) For the first time minute-hand and hour-hand will make 90° (or 15 points) difference in $\frac{60-15}{55} = \frac{45}{55}$ hours.

Since, in this case minute-hand goes till it appears to be 15 points behind of hour-hand (consider initially they are showing 12 O'clock) then you will see that at 12 : 49 : 05 two hand are making 90° angle between them, while it appears to be hour-hand is ahead of minute-hand.

Now, for every next time after $\frac{60}{55}$ hours they will show the same situation.

we have 12 hours.

So, $$\frac{45}{55}+\frac{60}{55}+\frac{60}{55}+\frac{60}{55}+\ldots = 12\,\text{h} = 12\times\frac{55}{55}$$

$$\Rightarrow \quad \frac{3}{11}+\frac{4}{11}+\frac{4}{11}\ldots = \frac{4\times 11}{11}$$

$$\Rightarrow \quad 3+4+4+\ldots = 44$$

$$\Rightarrow \quad 3+4n = 144$$

$\Rightarrow n = \frac{41}{4} = 10$, consider only integral value.

Thus, total $10 + 1 = 11$ times they will make 90° angle.

Hence, in 12 hours both hands make 90° angle $(11 + 11) = 22$ times in different positions.

Remember At 3 O'clock and 9 O'clock they are at right angled. Since, except between 2–4 O'clock and 8–10 O'clock in each hour both hands make 90° angle 4 times while in the 2–4 O'clock and 8–10 O'clock two hands makes three-three times in every two hour.

Exp. 4) Between 2 O'clock and 3 O'clock when two hands of a clock overlap each other?

Solution To overlap or overtake minute-hand has to reduce the gap of 10 points. Since at 2 O'clock two hands are 10 point apart.

$$\therefore \quad \text{Time} = \frac{\text{Distance advanced}}{\text{Relative speed}}$$

$$= \frac{10}{55}\,\text{h} = \frac{10}{55}\times 60\,\text{min} = 10\,\text{min}\ 54\,\text{s}$$

Thus, at 2 : 10 : 54 both hands of a clock coincide.

Exp. 5) Between 6 am and 7 am when the two hands of a clock coincide.

Solution $\text{Time} = \frac{30}{55}\,\text{h} = \frac{30}{55}\times 60\,\text{min} = 32\,\text{min}\ 43\,\text{s}$

(Distance advanced $= 6\times 5 = 30$ points)

Thus at 6 : 32 : 43 two hands of a clock coincide.

Exp. 6) Between 11 O'clock and 12 O'clock when will they coincide.

Solution Distance advanced at 11 O'clock = 55 points

Relative speed = 55 point/h

$$\therefore \quad \text{Time} = \frac{55}{55} = 1\,\text{h}$$

Hence, they will coincide at $(11 + 1) = 12$ O'clock.

Exp. 7) Between 3 O'clock and 4 O'clock when will the two hands make 36° angle between them:

(i) when hour-hand is ahead of minute-hand.

(ii) when minute-hand is ahead of hour-hand.

Solution

(i) 36° = 6 points

Now, at 3 O'clock two hands are separated by exactly 15 points to which we have to reduce upto 6 points. Thus, we have to reduce $15 - 6 = 9$ points distance, with the relative speed of 55 point/h.

$\therefore$ Time required $= \frac{9}{55}\,\text{h} = \frac{9}{55}\times 60 = 9\,\text{min}\ 49\,\text{s}$

Thus at 3 : 09 : 49 they are 36° apart from each other.

(ii) At 3 O'clock both hands are 15 points apart so to make them 6 points apart minute-hand has to move for $(15 + 6) = 21$ points, since minute-hand has to go 6 points ahead of hour hand, after crossing the hour-hand.

$$\therefore \quad \text{Time} = \frac{21}{55}\,\text{h} = \frac{21}{55}\times 60\,\text{min} = 22\,\text{min}\ 54\,\text{s}$$

Thust at 3 : 22 : 54, both hands will be 6 points (or 36°) apart from each other.

Did you notice something?

The same angle can be formed in two situations, one when hour-hand is ahead of minute-hand and when minute-hand is ahead of hour-hand. Thus, you can find the required time by dividing the required difference of points (which you have to either create or reduce) by the relative speed.

Exp. 8) What is the angle between the two hands at 3 : 10 am?

Solution Assume 60th point (*i.e.*, when it is 12 O'clock) as the origin.

***Step* 1.** Find the distance of minute-hand from the origin.

***Step* 2.** Find the distance of hour-hand from the origin.

***Step* 3.** Take the difference between two values obtained in step 1 and step 2.

$\therefore$ ***Step* 1.** 10 point = 60°

***Step* 2.** 90° + 5° = 95° (In 10 min hour-hand moves 5°)

***Step* 3.** 95° – 60° = 35°

Thus, at 3 : 10 am two hands are 35° apart.

Exp. 9) What is the angle between two hands of a clock at 7 : 35?

Solution ***Step* 1.** At 7 : 35, minute-hand is $35 \times 6 = 210°$ away from origin.

***Step* 2.** At 7 : 35 hour-hand is $7 \times 30 + 35 \times \frac{1}{2} = 210 + 17.5$

$= 227.5°$ away from the origin.

***Step* 3.** $227.5 - 210 = 17.5°$

Thus at 7 : 35, both hands make 17.5° angle between them.

NOTE A minute-hand moves 6° in one minute while a hour-hand moves $\left(\frac{1}{2}\right)°$ in one minute.

CAT-Test

Questions Helping you bell the CAT

LEVEL 01 > BASIC LEVEL EXERCISE

1 '*A*' goes 10 km distance with average speed of 6 km/h while rest 20 km he travels with an average speed of 15 km/h. What is the average speed of '*A*' during the whole journey?

(a) 10 km/h (b) 12 km/h

(c) 13 km/h (d) 14.5 km/h

2 *A* covers half of his distance with 20 km/h and rest with 30 km/h. What is the average speed during the whole journey?

(a) 20 km/h (b) 24 km/h

(c) 25 km/h (d) 26 km/h

3 *A* covers 1/3rd of his journey at the speed of 10 km/h and half of the rest at the speed of 20 km/h and rest at the speed of 30 km/h. What is the average speed of *A*?

(a) $6\frac{2}{11}$ km/h (b) $16\frac{4}{11}$ km/h

(c) $16\frac{4}{11}$ m/s (d) none of these

4 *A* covers 1/4th of his journey at 20 km/h and 1/3rd of the rest at 25 km/h and half of the rest at 30 km/h and rest at the speed of 40 km/h. What is the average speed of *A*?

(a) $13\frac{78}{89}$ km/h (b) 12 km/h

(c) $26\frac{86}{89}$ km/h (d) 28 km/h

5 *A* covered half of his journey at 20 km/h and rest at x km/h, then his average speed is 24 km/h. What is the value of x?

(a) 30 (b) 32

(c) 36 (d) 40

6 A man covered half of the distance at $3x$ km/h and rest at $5x$ km/h. What is the average speed of the man?

(a) $4x$ km/h (b) $3.5x$ km/h

(c) $3.75x$ km/h (d) none of these

7 A person goes to his office at 1/3rd of the speed at which he returns from his office. If the average speed during the whole trip (*i. e.*, one round) is 12 km/h. What is the speed of the person while he was going to his office?

(a) 10 (b) 6

(c) 8 (d) can't be determined

8 A person *X* starts from Lucknow and another persons *Y* starts from Kanpur to meet each other. Speed of *X* is 25 km/h, while speed of *Y* is 35 km/h. If the distance between Lucknow and Kanpur be 120 km and both *X* and *Y* start their journey at the same time, when will they meet?

(a) 1 h later (b) 2 h later

(c) $\frac{1}{2}$ h later (d) 3 h later